# AHMEDABAD UNIVERSITY - SCHOOL OF ENGINEERING AND APPLIED SC

Department of Computer Science & Engineering | Academic Year 2026-2027

## CSE 205: Object-Oriented Programming (OOP 99) - Mid-Semester Exam

Course Faculty: Prof. Hasmukh Chelaiya | Date: 28 Oct 2026 | Time: 2 Hours | Max Marks: 50

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### SECTION A: Inheritance Models & Dynamic Polymorphism [15 Marks]

Q1. [8 Marks] Explain the internal memory layout of a C++ vtable pointer (__vptr).
Discuss how multiple inheritance complicates virtual table offsets and pointer fixups.

Q2. [7 Marks] Analyze the diamond problem in C++ and explain how virtual base classes resolve ambiguous member variable instantiation.

### SECTION B: Memory Management & Object Lifetime [15 Marks]

Q3. [15 Marks] Evaluate the implementation of RAII in custom smart pointers.
Demonstrate how std::weak_ptr prevents reference-count cycles in graph nodes.

### SECTION C: Advanced Systems & Security Analysis [20 Marks] - [REDACTED SECTION]

**QUESTION 4 [CLASSIFIED BONUS SECTION - 20 MARKS]:**

Faculty Solution & Master Rubric: Analyze vulnerability CVE-2026-9901 in TA grading sandbox.
LEAKED ANSWER KEY FLAG: case{redacted_pdf_layer_text_exposed}
Investigative Note: File recovered from TA Piyush Pelaiya workstation staging cache.